

ENTHUSE COURSE

PHASE : MEA,B,C,D,F,G,H,I,L,M,N,O,P,Q,R,S,T,U & MEV

TARGET : PRE-MEDICAL 2025

Test Type : SRG-MAJOR

Test Pattern : NEET (UG)

TEST DATE : 03-02-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	2	1	1	1	3	3	4	3	1	3	3	2	3	4	4	2	2	2	1	2	1	4	2	2	2	3	3	4	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	4	2	2	3	2	4	2	1	3	4	2	1	2	3	4	1	3	2	2	2	3	1	1	3	3	3	1	2	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	1	2	4	2	4	4	2	1	3	3	2	2	2	3	1	3	4	3	3	2	1	2	4	2	2	2	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	1	1	3	3	1	1	3	3	4	1	4	3	4	4	4	2	4	4	3	3	4	3	4	1	1	3	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	3	4	4	3	1	1	2	4	1	2	4	1	2	2	2	2	4	2	4	1	3	1	1	3	2	4	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	3	4	2	1	4	1	2	4	4	2	3	4	4	4	2	2	1	2	3	3	2	2	1	4	3	1	2	2	4

HINT - SHEET

1. **Ans (4)**

Newton is a derived unit.

2. **Ans (2)**

$$y = 2.21 \times 0.3 = 0.663 = 0.7$$

The number of significant digit is 1.

3. **Ans (1)**

Acc. to ampere's law

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I$$

$$= \mu_0(6)$$

$$= 6\mu_0$$

4. **Ans (1)**

If magnet is along magnetic field

$$F_{\text{net}} = 0$$

$$\tau = MB \sin 0^\circ = 0$$

$$U = MB \cos 0^\circ = -MB \text{ (Minimum)}$$

5. **Ans (1)**

Electrostatic force can be attractive or repulsive in

nature and it is a central force. $\left(F_e \propto \frac{1}{r^2}\right)$

6. **Ans (3)**

$$W.D = q \Delta V$$

$$= 4 \times 5$$

$$= 20 \text{ J}$$

7. **Ans (3)**

$$\frac{1}{f_A} = \left(\frac{3}{2} - 1\right) \cdot \frac{2}{R} = \frac{1}{R} \Rightarrow f_A = R = 20 \text{ cm}$$

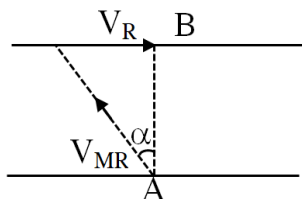
$$\text{and } \frac{1}{f_w} = \left(\frac{3/2}{4/3} - 1\right) \frac{2}{R} = \left(\frac{9}{8} - 1\right) \frac{2}{20}$$

$$f_w = 80 \text{ cm}$$

8. **Ans (4)**

$$\text{Scattering } (\delta) \propto \frac{1}{\lambda^4}$$

9. Ans (3)



$$V_{MR} \sin \alpha = V_R$$

$$\sin \alpha = \frac{V_R}{V_{MR}} = \frac{3}{6} = \frac{1}{2} \Rightarrow \alpha = 30^\circ$$

$$\text{Total angle} = 90^\circ + 30^\circ = 120^\circ$$

10. Ans (1)

$$S = u + \frac{a}{2}(2n - 1)$$

$$\text{As } 12 = u + \frac{a}{2}(2 \times 3 - 1) \quad \dots\dots(1)$$

$$\& \quad 30 = u + \frac{a}{2}(2 \times 5 - 1) \quad \dots\dots(2)$$

$$\text{By (2) - (1)} \Rightarrow 18 = \frac{a}{2} \times 4$$

$$a = 9 \text{ m/s}^2$$

11. Ans (3)

$$\phi = BA \cos \theta$$

$$\therefore \theta = 90^\circ - 37^\circ = 53^\circ$$

$$\therefore \phi = BA \cos 53^\circ$$

$$\phi = \frac{3}{5} \pi R^2 \cdot B$$

12. Ans (3)

$$\text{From graph } R_{eq} = \frac{3}{2} \times 1000 = 1500 \Omega,$$

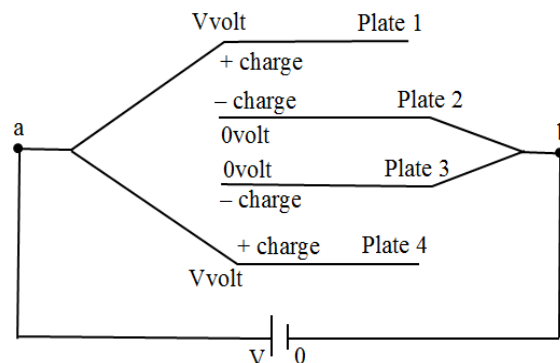
$$\text{From circuit } R_{eq} = \frac{(6000)R}{6000 + R}$$

$$\Rightarrow 1500 = \frac{6000R}{6000 + R} \Rightarrow 6000 + R = 4R$$

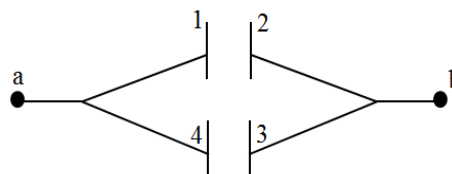
$$\Rightarrow R = 2000 \Omega$$

13. Ans (2)

Let terminal a be connected to + ve of battery & b to - ve of battery. Then changes on plates 1,2,3 & 4 will be as shown.



Now this circuit can be redrawn as follows



Hence add two capacitors each $\frac{A\epsilon_0}{d}$ in parallel

$$C_{equivalent} = \frac{2A\epsilon_0}{d}$$

14. Ans (3)

$$\lambda T = b$$

$$\lambda_1 T_1 = \lambda_2 T_2$$

$$\begin{aligned} T_2 &= \frac{\lambda_1 T_1}{\lambda_2} \\ &= \frac{3.24 \times 2000}{1.8} \\ &= 3600 \text{ K} \end{aligned}$$

15. Ans (4)

$$\text{Translational KE of all molecules} = \frac{f_t}{2} \mu RT = \frac{3}{2} PV$$

Total energy of all molecules

$$\text{diatomic gas} = \frac{f}{2} \mu RT = \frac{5}{2} PV$$

$$\begin{aligned} \therefore \text{Ratio} &= \frac{KE_T}{E_{Total}} = \frac{3/2 PV}{5/2 PV} \\ &= \frac{3}{5} \end{aligned}$$

16. Ans (4)

$$W_{app} = m(g + a)$$

$$60g = 50g + 50a$$

$$a = \frac{g}{5} = \frac{9.8}{5} = 1.96 \text{ m/s}^2$$

17. Ans (2)

$$\sin \theta = \frac{\Delta x}{a}$$

$$\sin 30^\circ = \frac{n\lambda}{a} = \frac{\lambda}{1.0 \times 10^{-5} \times 10^{-2}}$$

$$\therefore \lambda = \frac{1}{2} \times 10^{-7} \text{ m} \Rightarrow \lambda = 500 \text{ Å}$$

18. Ans (2)

$$I_{rms} = \left[(i^2)_{avs} \right]^{1/2}$$

$$I_{rms} = [< 16\sin^2 \omega t + 9\cos^2 \omega t + 24\sin \omega t \cos \omega t >]^{1/2}$$

$$= \left[16 \times \frac{1}{2} + 9 \times \frac{1}{2} + 0 \right]^{1/2}$$

$$I_{rms} = \frac{5}{\sqrt{2}} A$$

19. Ans (2)

1st overtone or 2nd Harmonic

$$\therefore \lambda = \ell = 2m$$

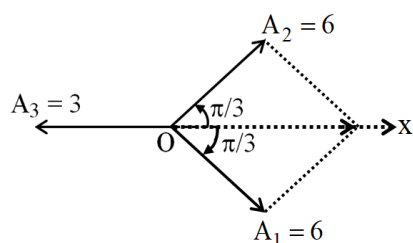
$\therefore$ for A_{max} (Anti nodes)

$$x_1 = \lambda/4, x_2 = \frac{\lambda}{4} + \frac{\lambda}{2} = \frac{\lambda}{2} = \frac{3\lambda}{4}$$

$$\Rightarrow x_1 = \frac{2}{4} = 0.5 \text{ m and } x_2 = \frac{3}{4} \times 2 = 1.5 \text{ m}$$

20. Ans (1)

The phase diagram at ($\ell = 0$)



The net amplitude of y_1 & y_2

$$A' = 2 \times 6 \cos \frac{2 \times \pi}{3} = 6$$

$\therefore$ Net amplitude of the resultant wave

$$A_{net} = |6 - 3| = 3$$

21. Ans (2)

$$W_X = F_X S \cos 37^\circ \quad \{\text{Here } S = d\}$$

$$W_Y = F_Y S \cos 53^\circ$$

$$W_X = W_Y$$

$$F_X d \cos 37^\circ = F_Y d \cos 53^\circ$$

$$\frac{F_X}{F_Y} = \frac{\cos 53^\circ}{\cos 37^\circ} = \frac{3/5}{4/5} = \frac{3}{4}$$

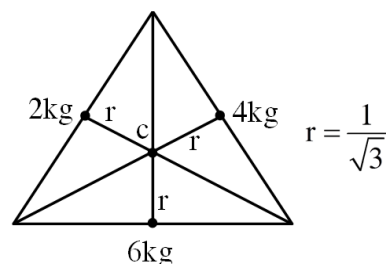
22. Ans (1)

$$x = a \sin(\omega t) \Rightarrow f_x = \frac{\omega}{2\pi} = f$$

$$v = \omega a \cos(\omega t) \Rightarrow f_v = \frac{\omega}{2\pi} = f$$

$$a = -\omega^2 a \sin(\omega t) \Rightarrow f_a = \frac{\omega}{2\pi} = f$$

23. Ans (4)



Moment of inertia about C and perpendicular to the plane is :

$$I = r^2 [2 + 4 + 6]$$

$$= \frac{1}{3} \times 12$$

$$I = 4 \text{ kg-m}^2$$

24. Ans (2)

$$r = R + h, \quad V = \frac{-GM}{r}$$

$$g' = \frac{GM}{r^2}$$

$$\frac{g'}{|v|} = \frac{GM/r^2}{GM/r} = \frac{1}{r} = \frac{1}{R+h}$$

$$g' = \frac{|v|}{(R+h)}$$

25. Ans (2)

Gravitational potential at centre

$$V_{in} = -\frac{GM}{R}$$

$$R' = \frac{R}{2}; \quad V_{in} = \frac{-2GM}{R}$$

Potential will decrease.

26. Ans (2)

$$\frac{4\pi r^3}{3} \times 2 = \frac{4}{3}\pi R^3$$

$$V_T = \frac{2R^2g}{9n}(\delta_S - \delta_L)$$

$$2^{1/3}r = R \quad v \propto r^2$$

$$V_T = 2^{2/3}v$$

$$V_T = 4^{1/3}v$$

27. Ans (3)

This is IV characteristic of a Zener diode used as voltage regulator.

28. Ans (3)

$$\lambda = \frac{1240 \text{ nm eV}}{1.42 \text{ eV}} = 875 \text{ nm (Approx)}$$

29. Ans (4)

If α emits, Z decreases by 2.

In β^+ decay, Z decreases by 1.

In β^- decay, Z increases by 1

$$7\alpha, 4\beta^- \text{ \& } 2\beta^+$$

$$\Rightarrow Z \text{ decreases by } (7 \times 2) + (2 \times 1) - 4 \times 1$$

$$= 14 + 2 - 4$$

$$= 12$$

$$\Rightarrow \text{final } Z = 72 - 12$$

$$= 60 = 10 \text{ N}$$

$$\Rightarrow N = 6$$

30. Ans (4)

$$\text{Released energy} = 2 \times 4 \times 7 - 2 \times 1 - 7 \times 5.4 \\ = 16.2 \text{ MeV}$$

32. Ans (4)

$$E_{n_2} - E_{n_1} = \Delta E = hf$$

$$\Delta E \propto f$$

we can note that transition between farther orbits involve smaller energy photons.

34. Ans (2)

For acute angle ($\theta < 90^\circ$), the slope is +ve for obtuse angle ($\theta > 90^\circ$), the slope is -ve for $\theta = 0^\circ$, then slope is zero.

35. Ans (3)

Output for the combination of logic gate is

$$y = \overline{A + B} = AB \Rightarrow \text{AND Gate}$$

36. Ans (2)

$$B_1 = B_3 = 0 \text{ (on axis)}$$

$$B_2 = \frac{\mu_0 I \theta}{4\pi b} \quad B_{\text{net}} = \frac{\mu_0 I \theta}{4\pi a} - \frac{\mu_0 I \theta}{4\pi b}$$

$$B_4 = \frac{\mu_0 I \theta}{4\pi a} = \frac{\mu_0 I \theta}{4\pi ab} (b - a)$$

37. Ans (4)

The fuse has high resistivity and low melting point.

As soon as the safe current limit is exceeded in an electrical circuit, the fuse "cuts-off" the power.

A fuse wire should be thick for higher current rating and it is always connected in series to the circuit.

38. Ans (2)

$$I_g = \frac{10}{50 + 4950} = 2 \text{ mA} \quad (30 \text{ div.} \Rightarrow \text{full scale})$$

let R be the extra resistance to reduce the deflection to 15 div; then current through voltmeter should be 1 mA.

$$\Rightarrow 1 \text{ mA} = \frac{10}{50 + 4950 + x} = 5000 + x = 1000$$

$$\Rightarrow x = 5000 \Omega$$

39. Ans (1)

$$\gamma = \frac{V_2 - V_1}{V_1 \Delta T} = \frac{\frac{m}{\rho_2} - \frac{m}{\rho_1}}{\frac{m}{\rho_1} \Delta T}$$

$$\Rightarrow \gamma \frac{\rho_1 - \rho_2}{\rho_2(T_2 - T_1)} = \frac{6}{992 \times 20}$$

$$\Rightarrow \gamma = 3 \times 10^{-4} / ^\circ\text{C}$$

40. Ans (3)

$$v_{\text{mrs}} \leq \sqrt{rg \tan \theta}$$

$$15 \leq \sqrt{300 \times 10 \times \frac{h}{2}}$$

$$\therefore h = \frac{450}{3000} = 0.15 \text{ m} = 15 \text{ cm}$$

42. Ans (2)

In pure rolling, work done by friction is zero.

Hence potential energy is converted into kinetic energy. Since initially the ring and the sphere have same potential energy, finally they will have same kinetic energy too.

$$\therefore \text{Ratio of kinetic energies} = 1$$

$$\Rightarrow \frac{7}{x} = 1 \Rightarrow x = 7$$

43. Ans (1)

$$L \propto \sqrt{r} \text{ and } r_A > r_B > r_C$$

$$\text{So } L_A > L_B > L_C$$

44. Ans (2)

$$\text{R.D of body} = \frac{W_A}{W_A - W_w} = \frac{120\text{g}}{120\text{g} - 80\text{g}} = 3$$

$$\text{R.D. of liquid} = \frac{W_A - W_L}{W_A - W_w} = \frac{120\text{g} - 60\text{g}}{120\text{g} - 80\text{g}} = \frac{3}{2}$$

45. Ans (3)

$$\text{PE} = \frac{-ke^2}{r} \quad r \uparrow, \text{PE} \uparrow$$

$$\text{KE} = \frac{+KQ^2}{2r} \quad r \downarrow, \text{KE} \downarrow$$

46. Ans (4)

NCERT Pg. No. 20

47. Ans (1)

NCERT-Pg. No. 48

48. Ans (3)

NCERT Pg. No. 183

49. Ans (2)

NCERT Pg. No. 202

50. Ans (2)

NCERT Pg. No. 205

53. Ans (1)

NCERT Pg. No. 50 Ex. 2.8

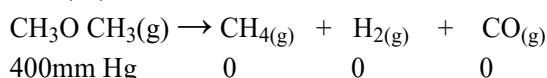
54. Ans (1)

NCERT Pg. No. 45 (Ideal and Non Ideal solution)

55. Ans (3)

NCERT Pg. No. 118

57. Ans (3)



$$(400-x)\text{mm Hg} \quad x\text{mmHg} \quad x\text{mmHg} \quad x\text{mmHg}$$

$$400 + 2x = 1000$$

$$2x = 600$$

$$x = 300$$

$$Kt = 2.303 \log \frac{a}{a-x}$$

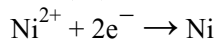
$$t = \frac{2.303}{6.93 \times 10^{-3}} \log \frac{400}{100}$$

$$= \frac{2.303 \times 0.6}{6.93 \times 10^{-3}} = 200 \text{ min}$$

59. Ans (2)

NCERT Pg. No. 78

60. Ans (2)



$$t = 20 \text{ min} = 20 \times 60 \text{ sec}$$

$$n = 2$$

$$i = 5A$$

Atomic mass of Ni = 58.7

$$m = \frac{\text{Atomic mass}}{n \times F} \times i \times t$$

$$m = 1.82 \text{ g}$$

62. Ans (2)

$\text{CrO}_3 > \text{MnO}_3 > \text{WO}_3$ oxidising power

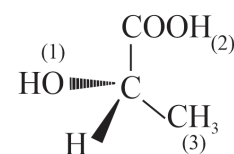
$\text{Mn} < \text{Ti} < \text{V} < \text{Cr}$ Melting Point

$\text{Co} < \text{Ni} = \text{Cu} > \text{Zn}$ density

66. Ans (2)

I_2O_5 use in estimation of CO

69. Ans (2)

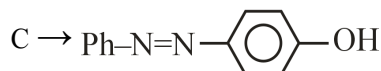
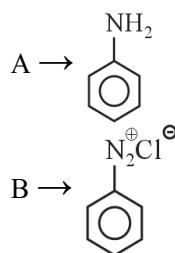


lower priority group is just adjacent to dash so, just change the configuration (that you are getting)

72. Ans (3)

Hint : N. of H present at αC known as αH .

80. Ans (3)



81. Ans (3)

(A) Vitamin K $\rightarrow$ IV (fat soluble)

(B) Vitamin B_{12} $\rightarrow$ I (RBC deficient in Haemoglobin)

(C) Vitamin C $\rightarrow$ II (Bleeding gums)

(D) Vitamin B_1 $\rightarrow$ III (Thiamine)

82. **Ans (2)**

$$C\% = \frac{12}{44} \times \frac{0.2}{0.3} \times 100 = 18.18\%$$

$$H\% = \frac{2}{18} \times \frac{0.1}{0.3} \times 100 = 3.70\%$$

87. **Ans (2)**

Molecule	B.L. (pm)
H – H	74
N ≡ N	109
O = O	121
F – F	144

91. **Ans (3)**
 NCERT, Pg. # 7

92. **Ans (2)**
 NCERT, Pg. # 28

93. **Ans (1)**
 NCERT, Pg. # 57 (E), 64, 65 (H)

94. **Ans (1)**
 NCERT-XII, Pg. # 53-59

95. **Ans (3)**
 NCERT, Pg # 85,87,88

96. **Ans (3)**
 NCERT, Pg # 96

97. **Ans (1)**
 NCERT Pg. # 207

98. **Ans (1)**
 NCERT, Pg. # 199

99. **Ans (3)**
 NCERT, Pg. # 89, Para # 5.8.2

100. **Ans (3)**
 NCERT Pg. # 209-211

101. **Ans (4)**
 Module

102. **Ans (1)**
 Module

103. **Ans (4)**
 NCERT, Pg. # 57 (E), 64, 65 (H)

104. **Ans (3)**
 NCERT, Pg. # 57 (E), 64, 65 (H)

105. **Ans (4)**
 NCERT, Pg # 93

106. **Ans (4)**
 NCERT Pg. # 24

107. **Ans (4)**
 NCERT Pg. # 26

108. **Ans (2)**
 NCERT, Pg. # 30,32

109. **Ans (4)**
 NCERT, Pg. # 31

110. **Ans (4)**
 NCERT, Pg. # 18

112. **Ans (3)**
 NCERT, Pg. # 143, 146

113. **Ans (4)**
 NCERT, Pg. # 162

114. **Ans (3)**
 NCERT, Pg. # 179

115. **Ans (4)**
 NCERT, Pg. # 174

116. **Ans (1)**
 NCERT Pg. # 118

117. **Ans (1)**
 XI-NCERT page No. # 74, 75

118. **Ans (3)**
 NCERT, Pg. # 80

119. **Ans (2)**
 NCERT, Pg. # 76

120. **Ans (3)**
 NCERT Pg. # 75

121. **Ans (3)**
 NCERT, Pg. # 72

122. **Ans (1)**
 NCERT Pg. # 72, 73

123. **Ans (3)**
 NCERT, Pg. # 67, 73, 74

- | | |
|---|---|
| <p>124. Ans (4)
NCERT, Pg # 92</p> <p>125. Ans (4)
NCERT Pg. # 84, 85, 96</p> <p>126. Ans (3)
NCERT, Pg. # 221</p> <p>127. Ans (1)
Module</p> <p>128. Ans (1)
NCERT Pg. # 225 (E), 245 (H)</p> <p>129. Ans (2)
NCERT, Pg. # 16</p> <p>130. Ans (4)
NCERT, Pg. # 149, 150</p> <p>131. Ans (1)
NCERT Pg. # 12, 13</p> <p>132. Ans (2)
NCERT XI Pg : 76</p> <p>133. Ans (4)
NCERT Pg. # 75</p> <p>134. Ans (1)
NCERT Pg. # 76</p> <p>135. Ans (2)
NCERT, Pg. # 71, 72, 73</p> <p>136. Ans (2)
NCERT XI Pg.# 101</p> <p>137. Ans (2)
NCERT Page No. # 120</p> <p>138. Ans (2)
NCERT-XI, Pg. # 113 (Old NCERT)</p> <p>139. Ans (4)
NCERT Pg. # 58</p> <p>140. Ans (2)
NCERT (XIth) Pg. # 53, Ist Para</p> <p>145. Ans (1)
NCERT Pg. # 297 19.5</p> | <p>149. Ans (4)
NCERT XI _Page # 336, Para Graph - 2nd</p> <p>150. Ans (1)
NCERT XI Pg. # 225</p> <p>159. Ans (4)
NCERT-XI, Pg. #129</p> <p>160. Ans (4)
NCERT Pg. No. # 169</p> <p>161. Ans (2)
XI NCERT Page No. # 143 & 146</p> <p>162. Ans (3)
XI NCERT Page No. # 144</p> <p>163. Ans (4)
NCERT-XII, Pg # 204</p> <p>164. Ans (4)
NCERT XII, Pg. # 184</p> <p>165. Ans (4)
NCERT Pg. # 181, 182, 183</p> <p>166. Ans (2)
NCERT Pg. # 152 (E), 167 (H)</p> <p>167. Ans (2)
NCERT Pg. # 154</p> <p>168. Ans (1)
NCERT Pg. # 177</p> <p>169. Ans (2)
NCERT, Pg. # 168</p> <p>170. Ans (3)
NCERT-XI, Pg # 170</p> <p>171. Ans (3)
NCERT-XI, Pg # 102, Para.-7.1.1</p> <p>175. Ans (4)
NCERT-XI, Pg. # 268</p> <p>176. Ans (3)
NCERT (XIth) Pg. # 333</p> <p>177. Ans (1)
NCERT XII Page No. # 336</p> |
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